SIMATS ENGINEERING
ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1 Hour
QUESTIONS: 5 **Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I Vasantha Nithi D Y (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause (□) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks
1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base
1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks
1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (□).
5. Explain each resolution step and write the final conclusion.

1.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Question 1 – Rain and Wet Ground

Problem Understanding

The knowledge base contains:

1. If it rains, the ground becomes wet.
2. It is raining.

The goal is to prove that **the ground is wet** using the **Resolution Algorithm**.

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1. Propositional Logic Representation

Let:

- **R** = It is raining
- **W** = Ground is wet

Rule:

R → W

Fact:

R

Goal:

W

---

2. CNF Conversion

Convert the implication:

R → W

Using:

P → Q = NOT P OR Q

Therefore:

NOT R OR W

The fact remains:

R

So the CNF clauses are:

- **NOT R OR W**

- **R**

---

3. Negate the Goal

Goal:

W

Negated goal:

NOT W

Add **NOT W** to the knowledge base.

---

4. Resolution Steps

Clause 1:

NOT R OR W

Clause 2:

R

Resolving them:

W

Now resolve:

W

with the negated goal:

NOT W

Therefore:

W OR NOT W → Empty Clause (□)

Resolution Tree

```
NOT R OR W       R
      \         /
       \       /
          W
          |
      NOT W
```

|

□

Result

The Empty Clause **(□)** is derived.

Therefore, the goal **W (ground is wet)** is successfully proved using Resolution.

Conclusion

Hence, using the Resolution Algorithm, it is logically proven that **the ground is wet** when it is raining.

Question 2 – Student Assignment Submission

Problem Understanding

The knowledge base states that if a student submits an assignment, the student receives internal marks. Rahul has submitted the assignment. The goal is to prove that **Rahul receives internal marks** using the Resolution Algorithm.

---

1. Propositional Logic Representation

Let:

- **S** = Rahul submitted the assignment
- **M** = Rahul receives internal marks

Rule:

S → M

Fact:

S

Goal:

M

---

2. CNF Conversion

Convert the implication:

S → M

Therefore:

NOT S OR M

Fact:

S

So the CNF clauses are:

- **NOT S OR M**
- **S**

---

3. Negate the Goal

Goal:

M

Negated goal:

NOT M

---

4. Resolution Steps

Resolve:

NOT S OR M

with:

S

This gives:

M

Now resolve:

M

with the negated goal:

NOT M

Therefore:

M OR NOT M → Empty Clause (□)

Resolution Tree

```
    NOT S OR M      S
```

```
\     /
 \   /
   M
   |
  NOT M
   |
   □
```

Result

The Empty Clause (□) is derived.

Therefore, **Rahul receives internal marks** is successfully proved using the Resolution Algorithm.

Conclusion

Hence, the given knowledge base logically proves that **Rahul receives internal marks**.

Question 3 – Library Membership

Problem Understanding

The knowledge base states that if a person is a library member, the person can borrow books. Priya is a library member. The goal is to prove that **Priya can borrow books** using the Resolution Algorithm.

---

1. Propositional Logic Representation

Let:

- **M** = Priya is a library member
- **B** = Priya can borrow books

Rule:

M → B

Fact:

M

Goal:

B

---

2. CNF Conversion

The rule:

M → B

becomes:

NOT M OR B

Fact:

M

So the CNF clauses are:

- **NOT M OR B**
- **M**

---

3. Negate the Goal

Goal:

B

Negated goal:

NOT B

---

4. Resolution

Resolve:

NOT M OR B

with:

M

Therefore:

B

Now resolve:

B

with:

NOT B

Therefore:

Empty Clause (□)

Resolution Tree

```
NOT M OR B      M
      \        /
       \      /
          B
          |
        NOT B
          |
          □
```

Final Conclusion

The Empty Clause **(□)** is obtained.

Therefore, **Priya can borrow books** is successfully proved using the Resolution Algorithm.

Question 4 – Placement Eligibility

Problem Understanding

The knowledge base states that if a student clears the aptitude test, the student is eligible for placement. Arun has cleared the aptitude test. The goal is to prove that **Arun is eligible for placement** using the Resolution Algorithm.

---

1. Propositional Logic Representation

Let:

- **A** = Arun cleared the aptitude test
- **E** = Arun is eligible for placement

Rule:

A → E

Fact:

A

Goal:

E

---

2. CNF Clauses

The rule:

A → E

becomes:

NOT A OR E

Fact:

A

Therefore, the clauses are:

- **NOT A OR E**
- **A**

---

3. Negate the Goal

Goal:

E

Negated goal:

NOT E

---

4. Resolution Steps

Resolve:

NOT A OR E

with:

A

Therefore:

E

Now resolve:

E

with:

NOT E

Therefore:

Empty Clause (□)

Resolution Tree

```
NOT A OR E      A
      \       /
       \     /
          E
          |
        NOT E
          |
          □
```

Result

The Empty Clause **(□)** is derived.

Therefore, **Arun is eligible for placement** is successfully proved.

Conclusion

Hence, the Resolution Algorithm proves that Arun is eligible for placement based on the given knowledge base.

Question 5 – Access Control System

Problem Understanding

The knowledge base contains two rules:

1. If a user enters the correct password, the user is authenticated.
2. If a user is authenticated, the user is granted access.
3. The user entered the correct password.

The goal is to prove that **the user is granted access** using the Resolution Algorithm.

---

1. Propositional Logic Representation

Let:

- **P** = User entered the correct password
- **A** = User is authenticated
- **G** = User is granted access

Rules:

P → A

A → G

Fact:

P

Goal:

G

---

2. CNF Conversion

Rule 1:

NOT P OR A

Rule 2:

NOT A OR G

Fact:

P

Therefore, the CNF clauses are:

- **NOT P OR A**
- **NOT A OR G**
- **P**

---

3. Negate the Goal

Goal:

G

Negated goal:

NOT G

---

4. Resolution Steps

First resolve:

NOT P OR A

with:

P

Therefore:

A

Next resolve:

NOT A OR G

with:

A

Therefore:

G

Finally resolve:

G

with:

NOT G

Therefore:

Empty Clause (□)

Resolution Tree

```
NOT P OR A     P
     \       /
        A
        |
   NOT A OR G
        |
        G
```

|

NOT G

|

□

Result

The Empty Clause **(**□**)** is derived.

Therefore, the user is **successfully granted access**.

Conclusion

Using the Resolution Algorithm, it is proved that the user is granted access because the correct password leads to authentication, which leads to access.